

Jayasri Suresh Vani

Software Engineer | Data Pipelines, Backend APIs & Applied AI/ML | Python, SQL | Purdue MS CS
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SUMMARY

MS Computer Science graduate (Purdue University) with 2+ years of combined experience across internships, research, and applied projects spanning full-stack development, backend engineering, AI/ML, and data engineering. Hands-on experience building RAG pipelines and LLM evaluation systems, full-stack web applications, and data-driven workflows. Delivered measurable impact across roles and projects — including a 45% reduction in reporting latency, 50% improvement in query performance, and 50% reduction in ML inference latency. Open to Software Engineer, Data Engineer and AI/ML Engineer roles.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript/JavaScript, Java

AI/ML & GenAI: RAG, LangChain, FAISS, LLM Evaluation, Prompt Engineering, Scikit-learn, Feature Engineering, NLP, Machine Learning

Backend & Web: FastAPI, REST APIs, React.js

Data & Cloud: ETL, Data Pipelines, PostgreSQL, GCP, Docker

EDUCATION

Purdue University | M.S., Computer Science (GPA: 3.9)

Aug 2024 - May 2026

Anna University | B.Tech, Information Technology (GPA: 4.0)

Oct 2020 - May 2024

EXPERIENCE

Community Dreams Foundation

Jul 2026

Generative AI Specialist

- Designed and deployed secure backend APIs to seamlessly connect production applications with Large Language Models (LLMs).
- Optimized data pipelines and model inference to significantly reduce latency and server infrastructure costs.
- Maintained core backend infrastructure, ensuring high availability, strict data security, and efficient system scaling.

Raven AI

Jul 2025 - May 2026

AI/ML Research & Development Intern

New York, NY

- Built Python-based evaluation pipelines transforming unstructured medical training dialogue into structured behavioral scores across 250+ conversational datasets.
- Benchmarked GPT-4o, Qwen3, and Llama using MAE, MSE, and R² metrics, automating scoring workflows to support model-selection decisions and speed up prompt experimentation.

Purdue University

Dec 2024 - Apr 2026

Technical Support Engineer → Lead Technical Support Engineer

- Resolved technical issues across OS, networking, authentication, and AV systems for 500+ daily users across classroom and lab environments.
- Re-imaged and configured 100+ lab systems and documented 20+ recurring fixes, improving setup consistency and reducing resolution time.

Techvolt Software Coimbatore

Feb 2023 - Nov 2023

Data Analyst & Machine Learning Intern

- Performed exploratory data analysis and data cleaning on 10,000+ records, improving data quality for downstream reporting and modeling.
- Optimized SQL queries and reporting workflows, reducing reporting latency by 45%, and built dashboards for stakeholder-facing analysis.
- Engineered features and trained classical ML models (Logistic Regression, Naive Bayes, Decision Trees), achieving 85%+ predictive accuracy.
- Compared multiple models and selected the best-performing approach using cross-validation and hyperparameter tuning.

Omega Binaries

Apr 2023 - May 2023

Software Developer Intern

- Developed a secure biometric authentication system using Java Swing, MVC architecture, and MySQL with encrypted credential storage.
- Implemented role-based access control and MFS100 fingerprint SDK integration for secure registration/login workflows.

PROJECTS

LumenDoc-QA | RAG Document Assistant

- Built a RAG pipeline with document ingestion, semantic chunking, embedding generation, and grounded answer generation for querying PDFs and web documents.
- Tested retrieval workflows across 50+ sample queries to evaluate answer relevance and source grounding.

Adaptive Learning Management System

- Built a role-based LMS supporting 500+ users with authentication, quiz workflows, reporting, and analytics dashboards.
- Optimized backend query retrieval time by 50% through database and dashboard workflow improvements.

AuctionEase | Online Auction Platform

- Built a real-time auction platform with live bidding, secure JWT authentication, and role-based access.
- Optimized SQL schemas and transaction workflows, reducing transaction latency by 40%.

NewsBias | Context-Aware Bias Detection

- Built a transformer-based NLP pipeline for political bias detection, achieving 92.78% classification accuracy.
- Used T5 for bias-aware rewriting, evaluated with ROUGE-1 score of 0.6053, and applied SHAP/LIME for model explainability.

Gesture2Globe | ASL Recognition & Translation

- Built a real-time computer vision pipeline for ASL recognition using hand-keypoint extraction and temporal modeling.
- Deployed a containerized inference backend on GCP Cloud Run, reducing inference latency by 50% through pipeline optimization.

Epileptic Seizure Classification

- Built a CNN/LSTM seizure classification model using EEG signal data and 2D signal transformations, achieving 92% classification accuracy.
- Applied wavelet transform and t-SNE visualization to improve model interpretability for biomedical signal reliability.